# Applied Econometrics Empirical Project

In this project, you will work in 4-to-5 person teams to come up with your own substantive research question and try to answer it with real data and the econometric methods you learn in this class. Your question needs to be feasible, and just as important, it needs to be a question you care about.

I suggest using one of these rich data sets that are available on the course web site along with documentation:

- The **High School Risk Survey** is composed of approximately 4,500 students from 10 Connecticut public high schools that were surveyed in 2006 about their attitudes toward risk and risky behaviors.

- The **Consumer Expenditure Survey (CE)** is collected every quarter by the US Bureau of Labor Statistics from a large nationally representative sample of American households. It includes numerous measures of income, purchases, household characteristics, and individual characteristics. This data is used by the government to compute poverty thresholds for social programs and revise the Consumer Price Index (CPI) market basket of goods and services. Businesses and academic researchers use the data to learn about consumer spending habits and trends. I've posted a large subset of the CE data collected/updated mid-year of 2024.

- The **General Social Survey (GSS)** has been collecting data on American knowledge, opinions, and behavior every two years since 1972. It asks many of the same questions every round giving us our best source of data for examining trends in these factors. It also asks new questions on timely issues every round. I've posted the data and documentation for the 2024 wave.

- Between 1995 and 1997, John McMillan and Christopher Woodruff surveyed **259 private manufacturing firms in Hanoi and Ho Chi Minh City, Vietnam**. They collected information about the firms, their top managers, their customers, and their suppliers.

With my permission, you may also use data you find yourself, though this is often a more difficult proposition. If you want to go this route, talk to me early in the process. You will **not** be allowed to conduct any experiment or survey that would require approval from a Human Subjects Review Board.

## Part 1: The Research Question and the Team Contract

You should write two to four paragraphs (about a page) about the question you hope to answer. Why is this question interesting? What do you think the answer might be and why? I encourage you to look at the documentation for the data sets listed above for ideas. Here are a few examples of feasible questions:

- How do characteristics of owners influence the revenue firms make?
- How does formal education affect people's knowledge of current events?
- What kinds of families have multiple cars?
- Does smoking affect grades in high school?
- What characteristics of countries determine how many gold medals they win at the Olympics?
- How do different NBA player performance statistics (e.g., points or rebounds per game) influence salaries?

There are three issues you may have to address in your research question:

1. **Causality vs. correlation:** Are you trying to estimate how changes in one thing (call this an *independent variable*) might affect another thing (call that a *dependent variable*)? This would be a statement about causation, and as many of you have already heard, correlation does not necessarily imply causation. It is much easier to describe how two things might be correlated, but it's much more interesting and useful to estimate causal effects.

2. **Time ordering:** In general, if you are estimating a causal effect, your determinants should be established before the measures of what they might be affecting. If you are looking at the effects of formal education on knowledge of current events, you should make sure formal education has been completed before you measure knowledge of current events. On the other hand, it doesn't make sense to look at how college education affects whether people smoke since most people who smoke start before they finish high school, and few people quit smoking in college.

3. **Confounders:** Are there variables that might influence both your independent and dependent variables and thus induce a correlation? e.g., People with high IQ's might be more likely to get more formal education and also keep up better with current events. People from poor families might be more likely to smoke and also receive poor grades in school. You will need to account for these potential *confounding variables* in your analysis.

The question you choose is not set in stone: You will be able to modify or change your research question later on if you need to do so.

At the same time that you work on your substantive research question, you will have to construct a "team contract" that will guide how you work together during the semester. See the Team Contract Template for details.

## Part 2: Data Description

Create your sample from your raw data. What are the relevant variables? There should be 10-15. How many observations do you have of each? You should have at least 100 observations and ideally quite a few more. Describe each variable's distribution using a tabulation if it is categorical or sample statistics and a histogram if it is continuous. Are your continuous variables normally distributed? At least one of your variables should be an *indicator variable* (only takes a value of 0 or 1). Describe the distribution of one of your continuous variables conditional on one of your indicators being zero and being one. Do not show histograms for categorical variables. Talk about the variables themselves rather than the name or number they might have in the survey (e.g., Don't talk about q55 but instead talk about income or grades or smoking status).

You need to be clear about your "unit of observation." That is, what does each observation correspond to? In a simple cross-sectional survey, the observations may be individuals (e.g., the General Social Survey and the High School Risk Survey), households (eg., the Consumer Expenditure Survey), firms (e.g., the Vietnamese firm data). Sometimes you will have observations of individuals across multiple years in which case we say our unit of observation is the "individual-year." You will also need to say what population your sample reasonably generalizes to. Is it all high school students in the US? Is it students who attended high school in Connecticut? Is it all firms in Vietnam?

Formally test differences in means for some of your variables in two subgroups. e.g., Are families with children more likely to own multiple cars than families without children? Do countries in colder regions of the world win more gold medals per capita at the Winter Olympics than countries in warmer regions? The results of these tests should give some insight into your research question and motivate the regression analysis you perform in part 3.

**Your report should be primarily English sentences with supporting tables and figures and should be no more than 7 pages.** It should include the title, research question, and motivation that you developed earlier, and you should incorporate any new ideas you've had since then and any feedback you received. Your tables should look pretty and should not simply be copied and pasted from Stata output. In addition to your report, you should create and hand in a single Stata .do file that performs all of the descriptive analysis.

*Please, see the grading rubric on the next page.*

**Your group's grade on the data description will be computed using the following rubric:**

| Rubric for Data Description | | |
|---|---|---|
| **Question (10 pts)** | | |
| | Clear (3 pts) | |
| | Important (4 pts) | |
| | Feasible (3 pts) | |
| | | |
| **Sample (15 pts)** | Explain source of data (5 pts) | |
| | Describe unit of observation (2 pts) | |
| | What population does the sample generalize to? | |
| | Number of observations (5 pts) | |
| | | |
| **Description of continuous variables (30 pts)** | (at least 3) | |
| | Include relevant variables (10 pts) | |
| | Means and standard deviations (7 pts) | |
| | Histograms (7 pts) | |
| | Clear English sentences (6 pts) | |
| | | |
| **Description of categorical variables (30 pts)** | | |
| | Include relevant variables (10 pts) | |
| | Tabulations (aka fractions of sample in each category (10 pts) | |
| | Clear English sentences (10 pts) | |
| | | |
| **Statistical Tests (15 pts)** | | |
| | Appropriate (10 pts) | |
| | Well explained (5 pts) | |
| **Final Grade** | | |

## Part 3: Data Analysis Plan

In 3-5 pages, you should first present and motivate your research question and then specify a set of regression models you plan to estimate. You should clearly define your dependent variables, explanatory variables of interest, and any control variables. You should explain how these models will help you answer your research question and why you have chosen these particular control variables. You should propose at least one "advanced" method from the third part of the course and explain why it is appropriate in this context.

Each individual will provide feedback to three other groups on their Data Analysis Plans and receive feedback from other groups (and the teaching team) on their group's Data Analysis Plan. You will not receive a grade on this part of the assignment, but you will incorporate this feedback when you implement your plan. Feedback will be composed of five categories:

1. **Research Question:** Is the question clear? Is it interesting?
2. **Dependent Variable(s):** Are they relevant to the research question? Are they reasonable measures of the outcome?
3. **Explanatory Variables:** Do they help answer the research question? Can you reasonably interpret coefficients on the research-relevant variables? Have important confounders been included?
4. **Advanced Method:** Is an advanced method proposed? Does it help answer the research question?
5. **Other:** Is there anything else you liked or worried about?

## Part 4: Data Analysis

Use an appropriate combination of multiple regression models and more advanced methods to answer your research question. You should estimate regression models with one to three different dependent variables and control for potential confounding variables. Carefully interpret the signs and magnitudes of coefficients in your preferred regression specifications. If there are additional potential confounders that you do not have in your data, discuss how omitting them may bias your results. Discuss potential intervening variables if you have any.

- Include the title, research question, and motivation you developed earlier.
- Include the equation for your preferred regression model specification.
- Include tables of regression results (including coefficients and standard errors) in your final report.
- **The maximum length for your report is 12 single spaced pages including tables and figures.** You will likely produce more during your research process, but it is important to know how to identify the most important results.

You should include the Stata code that performed all of your tests and regression analysis at the end of the PDF file you submit.

*Please, see the grading rubric on the next page.*

**Your group's grade on the Data Analysis will be computed using the following rubric:**

| Rubric for Data Analysis | | |
|---|---|---|
| **Question (10pts)** | | |
| | Clear (3 pts) | |
| | Important (4 pts) | |
| | Feasible (3 pts) | |
| | | |
| **Regression models (40pts)** | (at least 3) | |
| | Dependent make sense? (5 pts) | |
| | "Main" independent variables make sense? (5 pts) | |
| | Include potential confounders? Hope yes (5 pts) | |
| | Include big intervening variables? Hope no (5 pts) | |
| | Interpret signs and magnitudes (10 pts) | |
| | Interpret statistical significance (10 pts) | |
| | | |
| **More advanced methods (15pts)** | | |
| | Appropriate (10 pts) | |
| | Well explained (5 pts) | |
| | | |
| **Discussion/Conclusion (25 pts)** | | |
| | Main take away (10 pts) | |
| | Limitations of approach (10 pts) | |
| | Policy relevance (5 pts) | |
| | | |
| **Supplemental Material (10 pts)** | | |
| | .do file for analysis (10 pts) | |
| **Final Grade** | | |

## Individual Grades

Individual student grades on each part of the assignment will be the average of the group grade (detailed above) and a composite peer score. After the group submits a part of the assignment, each group member will anonymously rate each of their group peers on their dependability and effort. The peer score is composed of the ratings each student gets from their teammates. The goal of including a peer evaluation component in your grade is to reward students who participate in the project and discourage free-riding.